

IBVAP – Intelligent Border Video Analytics Platform

AI-Based Intelligent Video Analytics Platform for Border Surveillance using Existing CCTV Infrastructure

SIH Problem Statement Code: SIH26187

Organization: Ministry of Home Affairs

Category: Software

Theme: Smart Automation

1. One-Line Summary

IBVAP is an AI software layer that turns ordinary border CCTV cameras into a smart surveillance system that can automatically detect people, vehicles, faces, and number plates, identify intrusions, and send real-time alerts—without needing new "smart" hardware.

2. Problem Statement (In Simple Words)

- Border Security Forces use many CCTV cameras at Border Out Posts (BOPs), check posts, and border roads.
- Today, these cameras mostly just record video or show live feed.
- Security personnel must constantly watch screens to spot intrusions, suspicious movement, or specific persons/vehicles.
- Advanced features like facial recognition, ANPR (number plate reading), and intrusion detection usually need expensive proprietary hardware.
- This is costly and hard to deploy at scale, especially in remote border areas.

Goal: Build a cost-effective, software-only AI platform that adds intelligence to existing CCTV infrastructure and provides real-time, actionable alerts.

3. Proposed Solution (What We Are Building)

IBVAP is a software-defined video analytics platform that:

- Connects to existing IP CCTV cameras via their normal video streams (RTSP/HTTP).
- Runs AI models in real time to understand what is happening in the video.
- Detects and tracks:
 - People
 - Vehicles
- Performs:

- Face detection and matching against a watchlist
- Automatic Number Plate Recognition (ANPR)
- Virtual fence / zone intrusion detection
- Suspicious activity detection (e.g., loitering, night-time movement)
- Generates **real-time alerts** with snapshots/clips and logs all events for later search and analysis.
- Provides a **dashboard** for operators and APIs to integrate with existing command-and-control systems.

No special FRS/ANPR/smart cameras are needed—only standard CCTV plus our software.

4. Key Features

4.1 Core Capabilities

- **Human Detection & Tracking**
 - Detects persons in the camera view and assigns stable IDs over time.
- **Vehicle Detection & Classification**
 - Detects cars, trucks, buses, two-wheelers, etc., and tracks them.
- **Face Detection & Recognition**
 - Detects faces in the video.
 - Matches faces against a local "watchlist" (persons of interest).
- **Automatic Number Plate Recognition (ANPR)**
 - Detects vehicle number plates.
 - Reads plate text and logs it with time, camera, and image.
- **Virtual Fence / Intrusion Detection**
 - Operators can draw virtual lines or zones on each camera view.
 - System raises an alert when a person/vehicle crosses the line or enters a restricted zone.
- **Suspicious Activity Detection**
 - Loitering (staying too long in a sensitive area).
 - Night-time movement in restricted zones.
 - Rule-based patterns (e.g., repeated approaches to the fence).
- **Real-Time Alerting & Event Logging**
 - On-screen alerts for operators.
 - Optional external notifications (webhooks/SMS/email).
 - Full event log with filters (by time, camera, event type, plate text, face ID, etc.).

4.2 Operational Features

- **Dashboard for Operators**
 - Live camera grid with AI overlays (boxes, tracks).
 - Alert feed with thumbnails and quick actions.
 - Searchable event history.
- **Configuration UI**
 - Add/edit cameras (stream URL, location).
 - Draw virtual fences/zones visually.
 - Manage watchlists (faces, vehicles).
- **Integration**
 - REST/GraphQL APIs for external systems.
 - Webhooks to push alerts to existing MHA/BSF command systems.
- **Deployment Model**
 - Edge node per cluster of cameras (at BOP/sector HQ).
 - Optional central server for multi-BOP aggregation and long-term storage.
 - Works with intermittent connectivity; buffers locally when offline.

5. How It Works (Simple Architecture)

1. Cameras

- Existing IP CCTV cameras at border locations provide RTSP/HTTP video streams.

2. Edge Analytics Server (Our Software)

- Ingests multiple camera streams in parallel.
- Runs AI pipelines:
 - Detect & track people/vehicles
 - Detect & recognize faces
 - Detect & read number plates
 - Apply rules for intrusion, loitering, night movement
- Creates events and alerts, stores snapshots/clips.

3. Dashboard & Database

- Operators view live feeds and alerts on a web dashboard.
- All events are stored in a database for search, reporting, and integration.
- APIs allow other systems to fetch events or receive push notifications.

This matches the broader "Smart Border" vision of integrated command centres fusing multiple sensor feeds.

6. Technology Stack (High Level)

- **Language:** Python
- **Video Processing:** OpenCV, FFmpeg/GStreamer
- **AI Models:**
 - Person/Vehicle Detection: YOLOv8/YOLOv11
 - Tracking: ByteTrack / DeepSORT (or Ultralytics tracker)
 - Face Detection/Recognition: RetinaFace/SCRFD + InsightFace/ArcFace
 - ANPR: Plate detector (YOLO) + EasyOCR/PaddleOCR
- **Backend:** FastAPI (REST/WebSocket)
- **Frontend:** React/Next.js (or initial prototype in Streamlit/Gradio)
- **Database:** PostgreSQL (or SQLite for prototype)
- **Storage:** Local disk or S3-compatible storage for snapshots/clips
- **Deployment:** Docker + Docker Compose (easy deployment at remote sites)

All components are open-source and can run on modest hardware (with optional GPU for higher FPS).

7. Data Requirements (What Data We Use)

- **Live/Recorded Video Streams**
 - From existing CCTV (or simulated using public RTSP streams / recorded clips).
- **AI Model Datasets (for validation / fine-tuning)**
 - Person/vehicle: COCO, MOT, VisDrone, etc.
 - Faces: WIDER FACE, FDDB, plus a small custom watchlist.
 - Number plates: OpenALPR/CCPD-style datasets, plus Indian plate samples.
- **Operational Data (Managed by Our System)**
 - Camera registry (ID, location, stream URL).
 - Virtual fence/zone definitions per camera.
 - Watchlists (faces, vehicles).
 - User accounts and roles.
 - Event logs (time, camera, type, snapshot, metadata).
- **Test/Evaluation Data**
 - Held-out video clips with labeled events (intrusion, plates, faces) to measure accuracy and demonstrate performance.

8. Expected Impact & Benefits

- **Cost-Effective**
 - Uses existing CCTV; no need for expensive proprietary FRS/ANPR hardware.
- **Scalable**
 - Software can be deployed across many BOPs and check posts with similar hardware.
- **Improved Situational Awareness**
 - Automatic detection and alerting reduce dependence on continuous human monitoring.
 - Faster response to intrusions and suspicious activities.
- **Actionable Intelligence**
 - Searchable logs of people, vehicles, plates, and events.
 - Supports investigations and pattern analysis over time.
- **Aligned with National Initiatives**
 - Supports MHA's "Smart Border" and integrated command centre vision.

9. MVP Scope for Hackathon (What We Will Demonstrate)

For the SIH prototype, we will demonstrate:

- 2-4 simulated camera streams (using recorded videos / public streams).
- Real-time:
 - Person and vehicle detection + tracking
 - Virtual fence intrusion detection with alerts
- At least:
 - One working ANPR demo (reading a few number plates)
 - One working face recognition demo (small watchlist of a few people)
- Basic dashboard:
 - Live view with AI overlays
 - Alert list and simple event search
- Night-time movement detection logic (on low-light clips).

This covers all required capabilities from the problem statement in a working, end-to-end prototype.

10. Future Enhancements (Beyond Hackathon)

- Support for PTZ cameras and auto-tracking.
- More advanced behavior analytics (group formation, object carrying, etc.).
- Integration with thermal/IR cameras and drone feeds.
- Multi-site central command dashboard with map-based view of all BOPs.
- Hardening for production: high availability, role-based access control, audit logs, encryption, etc.

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